KULLIYYAH OF INFORMATION AND COMMUNICATION TECHNOLOGY

FINAL YEAR PROJECT 1 (FYP1)

PROJECT PROPOSAL

To be prepared BEFORE approaching a potential supervisor

Course Code	☐ CSCI 4401 ☐ BICS 4321
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Proposed Project Title	BagSense
Project Type	☐ Research Project ☐ System Development Project
Project Category / Area	AI

IMPORTANT

Maximum length: 6 pages. Complete all relevant sections before contacting a potential supervisor.

Do not list a supervisor's name until the staff member has agreed to supervise the project.

1. Project Overview

Provide a concise overview of the proposed project. Explain what you intend to study, investigate, design, or develop, and identify the intended users, context, or application area.

Project BagSense is an intelligent, energy-efficient Edge-IoT smart backpack system engineered to perform autonomous, non-line-of-sight inventory tracking inside bag's cavity. integrates a near-field Ultra-High Frequency (UHF) passive RFID reader with Hall-effect zipper-state sensing to dynamically inventory tagged personal items without requiring optical line of sight or manual unpacking. User interaction and glanceable inventory awareness are provided via an external chassis-mounted low-power E-paper display alongside a companion cross-platform Flutter mobile application connected over Bluetooth Low Energy (BLE) and Wi-Fi. The solution is targeted at university students and mobile professionals to eliminate gear misplacement and packing friction.

2. Background of the Study / Project

Explain the context and motivation for the project. For a research project, introduce the research area and why it is important. For a system development project, describe the problem or opportunity that motivates the proposed system.

University students, especially in IIUM, when they go across the KICT complex entails frequent transitions between lecture halls, laboratories, cafes, and residential areas can precipitate a pervasive problem like unintentional misplacement and omission of critical everyday gear that may eventually be a headache and burden. While existing consumer tracking accessories such as Apple AirTags address a single-item recovery, their dependence on individual active batteries and high budget resulting in full tagging of personal item were economically impractical. This is a clear operational opportunity to build an ambient, low-power smart container that automates item accountability at the point of packing.

3. Problem Statement / Problem Description

State the specific problem clearly. Explain the current limitation, gap, inefficiency, or unmet need. Avoid writing only a broad topic description; identify the actual problem that the project will address.

Existing approaches to personal item tracking suffer from a few critical structural weakness like :

1. High Per-Item Cost and Battery Dependance
    - Active Bluetooth beacon tags like AirTags and SmartTags cost between RM120 - RM RM160 per unit and rely on internal coin cells. Tagging ten common items within a backpack demands over RM1,200 solely in consumable tags.

2. Optical Occlusion in Closed Bags
    - Optical camera-based approaches inside bags fail due to extreme occlusions, lack of ambient illumination, and erratic item folding.

3. Passive and Friction-Heavy Verification

- Traditional packing requires manual physical searching or opening the bag. Users often only discover a missing item when they have already arrived at their lecture or examination venue.

4. Project Objectives

State 2-4 specific and achievable objectives. Use measurable action verbs where possible. The objectives should align directly with the problem stated above.

The core technical objective is to design, model, and deploy low-cost, production-feasible smart backpack that automates item awareness. The specific measurable objectives are :

1. Cavity Ergonomics and 3D Enclosure Modeling
    - To model the internal compartment liner, Faraday-damping lining, and modular electronic casing in Blender to ensure aesthetic discretion, weatherproofing, and user comfort.
2. Multi-Tag Non-Line-of-Sight Inventory Engine
    - To implement and optimize anti-collision UHF RFID interrogation capable of scanning and identifying up to 15 tagged items within the enclosed bag cavity in under 1.5 seconds with more than or equal to 98% recall.
3. Dual-Interface Visualization Framework
    - To develop a low-power external E-paper display for glanceable status outside the bag and a cross-platform mobile application using Flutter for interactive checklist creation, missing item push notifications, and schedule-based packing prompts.
4. Power-Aware Edge Firmware
    - To design an event-driven ESP32-S3 firmware architecture utilizing deep-sleep states and magnetic reed or hall-effect zipper switches to achieve an operational endurance exceeding 5 days per charge.
5. Field Validation in Campus Environment
    - To benchmark recognition accuracy, signal leakage boundaries, and usability across real-world testing scenarios within the KICT building.

5. Research Questions (Research Project Only)

For research projects, formulate questions that correspond to the research objectives. For system development projects, write 'Not Applicable'.

Not Applicable

6. Scope of the Study / Project

Define the boundaries of the project. State what will and will not be covered. For system development, include the target users, main functions, platform, data source, or deployment context where relevant.

1. Target Users

- Designed for university students, lecturers, and academic staff who routinely carry multiple study and technological items across campus venues.

2. Main Functions
    - Autonomous In-Cavity Inventory Scanning that identifies up to 15 passive RFID-tagged personal belongings without requiring optical line of sight in under 1.5 seconds.
    - Event-Driven Sensing that detects backpack opening and closing states automatically using magnetic Hall-effect zipper switches to initiate scanning routines.
    - Dual-Channel Status Output that renders quick and glanceable missing or present item indicators on an external E-paper panel while syncing itemized checklist logs to a smartphone.
    - Intelligent Power Management that enforces deep-sleep states between zipper events to prolong battery operation up to 5 days.
3. Platform and Hardware Specifications
    - Embedded Edge Controller: ESP32-S3 (Dual-Core 240 MHz, BLE 5.0 + Wi-Fi) running FreeRTOS tasks written in C++.
    - Scanning & Display Hardware: YRM100/MagicRF Nano UHF RFID reader module (865–868 MHz), compact 50 Omega near-field patch antenna, and a 2.9-inch SPI E-paper panel.
    - Mobile Application: Cross-platform front-end built using the Flutter framework (Dart) for Android and iOS.
    - Mechanical Chassis: Standard student multi-compartment backpack retrofit, customized with 3D-printed snap-fit protective enclosures modeled in Blender.
4. Data Source
    - Data is gathered directly from passive UHF RFID tag responses like Impinj Monza / Alien Higgs-3 transponders inside the bag cavity and Hall-effect sensor state transitions. Custom packing lists and user preferences are maintained via local storage in the Flutter mobile application.
5. Deployment Context
    - Testing and deployment are bounded to campus environments specifically within lecture halls, study areas, and laboratories across the KICT building.
6. Project Boundaries
    - No Global GPS Tracking where the system focuses strictly on inventory containment within the bag and does not incorporate real-time outdoor GPS or cellular positioning.
    - No Active Tracking per Item wherethe project does not utilize active LE/UWB tags such as Apple AirTags for individual items and tracking is strictly limited to passive, batteryless UHF RFID stickers.
    - No External Scanning or Confined Read Zone where The RFID field is shielded and damped with internal Faraday fabric to prevent reading tags carried by other individuals standing nearby outside the bag

7. Significance / Expected Contribution

Explain why the project is useful and who may benefit. State the expected contribution to users, organisations, the discipline, or future research.

1. Cost efficiency where the project drastically lowers the cost of item-level tracking by utilizing passive RFID transponders costing less than RM1.00 per tag, compared to RM120+ active beacons.
2. Frictionless Usability where the project automates item verification through zipper open/close events, eliminating manual unpacking or smartphone dependency via external glanceable E-paper feedback.

3. Academic and Engineering Contribution where the project demonstrates a practical edge implementation of near-field UHF RFID physics inside wearable, flexible containers, addressing electromagnetic boundary containment and low-power battery management.

8. Brief Review of Related Work

Summarise several relevant studies, systems, applications, methods, or technologies. Highlight what already exists and identify the gap or limitation that motivates your proposed project. Cite the sources properly.

1. Smart Luggage Tracking Using IoT and GPS (Jain et al., 2023): Explored smart luggage systems utilizing microcontrollers, GPS modules, and GSM/cellular telemetry to trace missing baggage. While effective for macroscopic bag location tracking, their solution does not monitor internal contents or detect individual missing items within the compartment.
2. Energy-Aware Task Scheduling for Batteryless/Constrained IoT (Sabovic et al., 2022): Addressed energy harvesting and power-state transitions in edge-constrained IoT devices. This provides the foundational rationale for BagSense's event-driven power architecture, which pairs deep-sleep modes with mechanical/magnetic zipper state triggers to limit active battery consumption.
3. Commercial Proximity Trackers (Apple AirTag, Samsung SmartTag): Leverage BLE and Ultra-Wideband (UWB) for peer-to-peer localization. However, each tracked entity requires dedicated power supplies and costly circuitry, making multi-item tracking inside a single personal bag economically prohibitive.

9. Proposed Methodology / Development Approach

Research project: briefly describe the research design, population/sample (if relevant), data collection, analysis method, and ethical considerations. System development project: describe the proposed development approach (e.g., Agile/Waterfall), requirements gathering, design, implementation, database (if applicable), and testing/evaluation.

The project follows an Agile / Prototyping development methodology across four key phases:

1. Requirements Gathering & Hardware Dimensioning: Define spatial constraints of standard campus backpacks, map tag read zones, and finalize power requirements and component selection.
2. CAD & Enclosure Design: Design snap-fit electronic enclosures, cable conduits, and the external bezel in Blender, followed by 3D printing and test fitment with Faraday attenuation mesh.
3. Firmware & Software Implementation:
    - Develop FreeRTOS tasks on ESP32-S3 for UHF reader communication, Hall-effect interrupt handling, E-paper SPI screen rendering, and BLE GATT server synchronization.
    - Develop the Flutter mobile companion app to configure inventory checklists and receive event logs.
4. Integration & Evaluation: Benchmark tag identification speed, evaluate false-negative rates with multiple tagged items, verify signal confinement within the chassis, and validate battery endurance across typical daily campus routines.

10. Expected Output / Deliverables

State the main outputs expected at the end of the project, such as a prototype, mobile/web application, trained model, dataset, analytical findings, dashboard, framework, or evaluation results.

1. Fully Functional BagSense Prototype: A retrofitted smart backpack integrated with an ESP32-S3 controller, near-field UHF RFID module, internal antenna, magnetic zipper sensors, and power management circuit.
2. Glanceable Display & Bezel Assembly: A 3D-printed outer chassis bezel mounting a 2.9-inch SPI E-paper status screen showing immediate bag inventory status.
3. Companion Flutter Mobile App: A mobile application running on Android/iOS supporting custom packing lists, category tags, and BLE sync.
4. System Firmware & Evaluation Report: Complete ESP32-S3 firmware codebase alongside a benchmark evaluation report validating reading accuracy (≥98%), latency (≤1.5), and multi-day battery endurance.

Student Readiness Checklist

☐ I have earned at least 75 credit hours

☐ I have passed prerequisite courses as follows:

- BICS 2302 - Data Structures and Algorithms
- BICS 2303-Intelligence System

Note: This proposal is intended for initial supervisor consideration and may be refined after discussion with the appointed supervisor.